Week 07 - Tutorial 01

Transcript Language

English

Hello everyone. In this lecture, in this tutorial video we will talk about similar matrices. So,

let us take an example. So, we start with some linear transformation T of x, y, z

is x, x plus y and x plus z. So, this is a linear transformation, from R3 to R3 and now we consider

ordered basis, so this is beta 1, this is the standard ordered basis we are considering 1, 0, 0;

0, 1, 0 and 0, 0, 1. So, this is the standard ordered basis.

Let us consider another ordered basis beta 2, let us take beta 2 as this 1, 0, 1;

0, 1, 1 and 1, 1, 0. So, if we write the matrix representation with respect to both the ordered

basis, so T with respect to beta 1, and for the domain and codomain in both places if we consider

the basis to be beta 1, then this will give us the matrix. So, let us write it T of 1, 0, 0

that is nothing but 1 comma 1 comma 1; T of 0, 1, 0 that is the matrix, so 0, 1, 0

and T of 0, 0, 1 this will give us 0, 0, 1.

So, the matrix we got here, so this matrix is nothing but first column will be 1, 1,

1; second column will be 0, 1, 0 and the third column will be 0, 0, 1.

So, this is the matrix representation with respect to standard ordered basis. Now, similarly if we

calculate T of these vectors in beta 2 and we write, we then get T of 1, 0, 1 is nothing but

1, 1, 2. T of 0, 1, 1 is nothing but 0, 1, 1 and T of 1, 1, 0 is 1, 2, 1.

Now, 1, 1, 2 we can write, express in terms in a linear combination of vectors in beta 2.

So, we will get 1 into 1, 0, 1 plus 1 into 0, 1, 1 plus 0 into 1, 1, 0. Similarly, we

can express the other two also, so the matrix we get is T beta 2 is given by 1, 1, 0; 0, 1, 0 and

0, 1, 1. So, I want all of you to check this representation. You can express all the images

these vectors and these vectors as a linear combination of vectors in beta 2

and then write the matrix representation. So, we get T beta 1 and T beta 2,

those are the matrix representation with respect to beta 1 and beta 2 respectively.

Now, these two matrix are similar, but for similar matrix we know that there should exists some,

so let us denote this matrix as A for removing the complication of notation and this matrix as B.

So, for as A and B are similar matrix, so there should exists some P such that P B P inverse

is A or even you can write A is nothing but P inverse, sorry B is nothing but P inverse AP.

So either of these, basically I just change the size, sides of P. So, we want to find this P.

What will be this P? So, what we got till now, till now we got that if we consider beta 1 for both domain and codomain we get the

matrix A as the representation of linear transformation. Similarly, for beta 2,

we got the matrix B. So, now if we represent this beta 2 in terms of beta 1, then we get

the change of basis matrix that will give us the matrix P. So, what we have to do? We have to

do, we have to express the vectors of beta 2 in terms of vectors of beta 1.

So, here beta 2 is nothing but 1, 0, beta 2 is nothing but 1, 0, 1; 0, 1, 1 and

1, 1, 0. Now we have to express this in terms of beta 1, beta 1 is our standard ordered basis.

So, this is 0, 1, 0 and 0, 0, 1. So, this is the standard ordered basis. So, we have to express,

we have to express beta 2 in terms of

beta 1. So, this is the matrix which we get in this process, we will is called as

change of basis matrix. So, let us try to write 1, 0, 1 in terms of

standard ordered basis so which is nothing but 1 into the first one 1, 0, 0; 0 into the second one

0, 1, 0 and 1 into the third one 0, 0, 1. So, similarly for the other two vectors we

can write this expression, so this is nothing but the matrix which we get by taking the vectors of

beta 2 as the columns of this matrix. So, the matrix we get is the column of this,

we replace the columns with the vectors in beta 2. So,

this is the matrix we get, this is the change of basis matrix and we denote it by P.

Now, if we calculate P of B we will get 1, 0, 1; 1, 1, 2 and 2, 1,

1. So, this is our P of B. And if we calculate A into P so this is the matrix P into B

and A into P will give us the same matrix 1, 0, 1; 1, 1, 2; 2, 1, 1. I want you to check verify this

two, these two matrices PB and AP. So, what we get here? We get PB is equal to AP. So, B equal to P

inverse AP. So, we can verify that this A and B are really a similar matrix, but P is constructed

by expressing beta 2 in terms of beta 1. Now, in the same example we have not changed

the linear transformation and we have also taken the beta 2, the basis as 1, 0, 1; 0, 1, 1 and 1, 1, 0 as earlier. So, this beta, B matrix we have calculated earlier.

Similarly, if we take the another basis beta 3 as 1, 0, 0; 1, 1, 0 and 1, 1, 1

you can calculate the matrix representation, you can verify it that it will be C.

Now, in this case we have also got two matrices B and C by just changing the basis. So,

this B and C will be similar and we want to find, we want to find some matrix Q such that QC Q

inverse will be B. That means, C will be Q inverse BQ. We want to find this Q. Now, how we can find

this Q? We have to express beta 3, the vectors in beta 3 in terms of the vectors in beta 2. We have

to find this matrix and that will be our Q. So, let us try to write the first vector 1, 0, 0

in terms of the vectors of beta 2. So, that is nothing but a into this

plus b into 0, 1, 1 and c into 1, 1, 0. So, what we get here? We get a plus c

and this b plus c and a plus b. So, our a plus b will be 1; b plus c will be 0

and a plus, sorry the first one is a plus c, a plus c will be 1, and the last one is a plus b that is 0. So, we got, we get 3 equations and if we solve it we will get a to be half,

b to be minus half and c to be half. So, similarly we can express the other two vectors

which are 1, 1, 0 and 1, 1, 1 in terms of vectors in beta 2 and if we write all these

things in the columns, as the coefficients as the columns of a matrix, then we get the matrix Q.

And I want all of you to verify that this is nothing but half, minus half, half which we

have already checked and the next one will be 0, 0, 1 and the last one will be half, half and half.

So, this is our matrix Q. Now if you calculate BQ, you will get half, 0, half;

half, 1, 3 by 2 and half, 1, half. So, this is the matrix BQ and which will be same as QC. So, if you

take product of Q and C and B and Q, you will get the same matrix.

So, we will get C equal to Q inverse BQ, where Q is nothing but the change of basis matrix.

So, how we can get Q? We just express all the vectors in beta 3

in terms of the vectors in beta 2. I will write this coefficient, this a, b and c for each case.

So, write a, b and c as the column of the matrix of Q. So, that is how we can calculate

Q and that is how we get, you can verify that two matrices are similar matrix or not. So, whenever

there is a change of basis and we get two representation of a linear transformation

with respect to that basis, then those two matrices are similar matrices. Thank you.